MTH 204 : Lecture 15

Nonhomogeneous linear system of ODEs :

A nonhomogeneous linear system of ODEs can be given by :

$$Y' = A(t)Y + g(t) \quad \ldots\ldots (1)$$

where the vector $g(t)$ is not identically zero.

If the entries of the matrix A and g vector are continuous on some interval J of the t-axis, then the general solution of (1) can be expressed as: $\quad Y = Y_h + Y_p$

where Y_h is the general solution of the associated homogeneous system of ODEs and Y_p is a particular solution of (1) on J

Method of Undetermined coefficients :

- This method is applied when A is a constant matrix and g is a sum of constant, powers, exponentials, Sine/Cosine functions.

Ex:

$$Y' = \begin{pmatrix} 1 & -2 \\ 3 & -4 \end{pmatrix} Y + \begin{pmatrix} -\sin t \\ -\cos t \end{pmatrix}$$

- First let us find the general solution of the homogeneous system.

For $A = \begin{pmatrix} 1 & -2 \\ 3 & -4 \end{pmatrix}$,

$$\det(A-\lambda I) = (1-\lambda)(-4-\lambda) - 3(-2) = \lambda^2+4\lambda-\lambda-4+6$$
$$= \lambda^2+3\lambda+2 = (\lambda+2)(\lambda+1)$$

So, the eigen values are $-2, -1$

For eigen vector corresponding to the eigen value $\lambda=-2$,

$$\begin{bmatrix} 1+2 & -2 \\ 3 & -4+2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow \left.\begin{aligned} 3x_1 - 2x_2 = 0 \\ 3x_1 - 2x_2 = 0 \end{aligned}\right\} \Rightarrow x_2 = \frac{3}{2}x_1 = 1.5x_1$$

So, $\begin{pmatrix} 1 \\ 1.5 \end{pmatrix}$ is an eigen vector corresponding to $\lambda = -2$

Now for eigen vector corresponding to the eigen value $\lambda = -1$

$$\begin{bmatrix} 1+1 & -2 \\ 3 & -4+1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow \left.\begin{aligned} 2x_1 - 2x_2 &= 0 \\ 3x_1 - 3x_3 &= 0 \end{aligned}\right\} \Rightarrow x_1 = x_2$$

So, $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$ is an eigen vector of A

corresponding to the eigen value -1

So, $Y_h = c_1 \begin{pmatrix} 1 \\ 1.5 \end{pmatrix} e^{-2t} + c_2 \begin{pmatrix} 1 \\ 1 \end{pmatrix} e^{-t}$

Since the RHS has $\begin{pmatrix} -\sin t \\ -\cos t \end{pmatrix}$, using method of undetermined coefficient we try a particular solution of the form

$$Y_p = \begin{pmatrix} a\cos t + b\sin t \\ c\cos t + d\sin t \end{pmatrix} = \begin{pmatrix} a \\ c \end{pmatrix} \cos t + \begin{pmatrix} b \\ d \end{pmatrix} \sin t$$

Then $\quad Y_p' = \begin{pmatrix} -a\sin t + b\cos t \\ -c\sin t + d\cos t \end{pmatrix} = -\begin{pmatrix} a \\ c \end{pmatrix}\sin t + \begin{pmatrix} b \\ d \end{pmatrix}\cos t$

Substituting in the ODE,

$$-\begin{pmatrix} a \\ c \end{pmatrix}\sin t + \begin{pmatrix} b \\ d \end{pmatrix}\cos t = A\left[\begin{pmatrix} a \\ c \end{pmatrix}\cos t + \begin{pmatrix} b \\ d \end{pmatrix}\sin t\right] + \begin{pmatrix} -\sin t \\ -\cos t \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} b \\ d \end{pmatrix} = A\begin{pmatrix} a \\ c \end{pmatrix} + \begin{pmatrix} 0 \\ -1 \end{pmatrix} = \begin{pmatrix} 1 & -2 \\ 3 & -4 \end{pmatrix}\begin{pmatrix} a \\ c \end{pmatrix} + \begin{pmatrix} 0 \\ -1 \end{pmatrix} \quad \ldots\ldots (1)$$

and $$-\begin{pmatrix} a \\ c \end{pmatrix} = A\begin{pmatrix} b \\ d \end{pmatrix} + \begin{pmatrix} -1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 & -2 \\ 3 & -4 \end{pmatrix}\begin{pmatrix} b \\ d \end{pmatrix} + \begin{pmatrix} -1 \\ 0 \end{pmatrix} \quad \ldots\ldots (2)$$

Substituting (1) in (2) we get

$$-\begin{pmatrix} a \\ c \end{pmatrix} = \begin{pmatrix} 1 & -2 \\ 3 & -4 \end{pmatrix}\left[\begin{pmatrix} 1 & -2 \\ 3 & -4 \end{pmatrix}\begin{pmatrix} a \\ c \end{pmatrix} + \begin{pmatrix} 0 \\ -1 \end{pmatrix}\right] + \begin{pmatrix} -1 \\ 0 \end{pmatrix}$$

$$\Rightarrow -\begin{pmatrix} a \\ c \end{pmatrix} = \begin{pmatrix} 1 & -2 \\ 3 & -4 \end{pmatrix}^2\begin{pmatrix} a \\ c \end{pmatrix} + \begin{pmatrix} 1 & -2 \\ 3 & -4 \end{pmatrix}\begin{pmatrix} 0 \\ -1 \end{pmatrix} + \begin{pmatrix} -1 \\ 0 \end{pmatrix}$$

$$\Rightarrow -\begin{pmatrix} a \\ c \end{pmatrix} = \begin{pmatrix} -5 & 6 \\ -9 & 10 \end{pmatrix}\begin{pmatrix} a \\ c \end{pmatrix} + \begin{pmatrix} 2 \\ 4 \end{pmatrix} + \begin{pmatrix} -1 \\ 0 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} -4 & 6 \\ -9 & 11 \end{pmatrix}\begin{pmatrix} a \\ c \end{pmatrix} = \begin{pmatrix} -1 \\ -4 \end{pmatrix} \Rightarrow \begin{pmatrix} a \\ c \end{pmatrix} = \begin{pmatrix} -4 & 6 \\ -9 & 11 \end{pmatrix}^{-1}\begin{pmatrix} -1 \\ -4 \end{pmatrix}$$

$$= \frac{1}{10}\begin{pmatrix} 11 & -6 \\ 9 & -4 \end{pmatrix}\begin{pmatrix} -1 \\ -4 \end{pmatrix} = \frac{1}{10}\begin{pmatrix} 13 \\ 7 \end{pmatrix} = \begin{pmatrix} \frac{13}{10} \\ \frac{7}{10} \end{pmatrix}$$

Then $\begin{pmatrix} b \\ d \end{pmatrix} = \begin{pmatrix} 1 & -2 \\ 3 & -4 \end{pmatrix} \begin{pmatrix} \frac{13}{10} \\ \frac{7}{10} \end{pmatrix} + \begin{pmatrix} 0 \\ -1 \end{pmatrix}$ (from ①)

$$\Rightarrow \begin{pmatrix} b \\ d \end{pmatrix} = \begin{pmatrix} -\frac{1}{10} \\ \frac{11}{10} \end{pmatrix} + \begin{pmatrix} 0 \\ -1 \end{pmatrix} = \begin{pmatrix} -\frac{1}{10} \\ \frac{1}{10} \end{pmatrix}$$

Therefore the solution is

$$a = \frac{13}{10} = 1.3, \quad b = -\frac{1}{10} = -.1$$

$$c = \frac{7}{10} = .7, \quad d = \frac{1}{10} = .1$$

Therefore the particular solution

$$Y_p = \begin{pmatrix} 1.3 \cos t - .1 \sin t \\ .7 \cos t + .1 \sin t \end{pmatrix}$$

Hence the general solution is

$$Y = Y_h + Y_p$$

$$= c_1 \begin{pmatrix} 1 \\ 1.5 \end{pmatrix} e^{-2t} + c_2 \begin{pmatrix} 1 \\ 1 \end{pmatrix} e^{-t} + \begin{pmatrix} 1.3 \cos t - .1 \sin t \\ .7 \cos t + .1 \sin t \end{pmatrix}$$

or equivalently

$$\left.\begin{aligned} y_1 &= c_1 e^{-2t} + c_2 e^{-t} + 1.3 \cos t - .1 \sin t \\ \text{and } y_2 &= 1.5 c_1 e^{-2t} + c_2 e^{-t} + .7 \cos t + .1 \sin t \end{aligned}\right\}$$

Exercise: $Y' = \begin{pmatrix} -3 & 1 \\ 1 & -3 \end{pmatrix} Y + \begin{pmatrix} -6 \\ 2 \end{pmatrix} e^{-2t}$

Method of Variation of Parameters

This method can be applied to nonhomogeneous linear system $Y' = A(t)Y + g(t)$ for nonconstant matrix $A(t)$ and general g.

It yields a particular solution Y_p on some open interval J on the t-axis if the general solution of the associated homogeneous system $Y' = A(t)Y$ is known.

If the general solution of the associated homogeneous system is of the form

$$Y_h = (Y_1 \; Y_2 \cdots Y_n)\,\mathbf{c} = Y(t)\,\mathbf{c}$$

then we will look for a solution of the form

$$Y_p = Y(t)\,u(t)$$

Then $Y_p' = A(t)\,Y(t)\,u(t) + g(t)$

$\Rightarrow \quad Y'(t)\,u(t) + Y(t)\,u'(t) = A(t)\,Y(t)\,u(t) + g(t)$

$\Rightarrow \quad Y'u + Yu' = AYu + g$

Since the columns of Y are solutions of the associated homogeneous system,

we have $\quad Y' = AY$

Then $\quad Yu' = g$

$$\Rightarrow \quad \boxed{u' = Y^{-1} g}$$

Ex:

$$Y' = \begin{pmatrix} 1 & -2 \\ 3 & -4 \end{pmatrix} Y + \begin{pmatrix} -\sin t \\ -\cos t \end{pmatrix}$$

In the previous example, we have calculated the general solution of the associated homogeneous system:

$$Y_h = c_1 \begin{pmatrix} 1 \\ 1.5 \end{pmatrix} e^{-2t} + c_2 \begin{pmatrix} 1 \\ 1 \end{pmatrix} e^{-t}$$

$$= \begin{pmatrix} e^{-2t} & e^{-t} \\ 1.5e^{-2t} & e^{-t} \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} = \boxed{YC}$$

So, $Y_1 = \begin{pmatrix} e^{-2t} \\ 1.5\,e^{-2t} \end{pmatrix} \quad Y_2 = \begin{pmatrix} e^{-t} \\ e^{-t} \end{pmatrix}$

$$Y^{-1} = \frac{1}{-.5\,e^{-3t}} \begin{pmatrix} e^{-t} & -e^{-t} \\ -1.5e^{-2t} & e^{-2t} \end{pmatrix}$$

$$= -2 \begin{pmatrix} e^{2t} & -e^{2t} \\ -1.5\,e^{t} & e^{t} \end{pmatrix}$$

So, $u' = Y^{-1} g = -2 \begin{pmatrix} e^{2t} & -e^{2t} \\ -1.5e^{t} & e^{t} \end{pmatrix} \begin{pmatrix} -\sin t \\ -\cos t \end{pmatrix}$

$$= -2 \begin{pmatrix} -e^{2t}\sin t + e^{2t}\cos t \\ 1.5e^{t}\sin t - e^{t}\cos t \end{pmatrix} = \begin{pmatrix} 2e^{2t}(\sin t - \cos t) \\ e^{t}(-3\sin t + 2\cos t) \end{pmatrix}$$

Then $u = \int \begin{pmatrix} 2e^{2t}(\sin t - \cos t) \\ e^{t}(-3\sin t + 2\cos t) \end{pmatrix} dt = \begin{pmatrix} \frac{2e^{2t}}{5}(\sin t - 3\cos t) \\ \frac{e^{t}}{2}(5\cos t - \sin t) \end{pmatrix}$

Then $Y_p = Y u = \begin{pmatrix} e^{-2t} & e^{-t} \\ 1.5e^{-2t} & e^{-t} \end{pmatrix} \begin{pmatrix} \frac{2e^{2t}}{5}(\sin t - 3\cos t) \\ \frac{e^{t}}{2}(5\cos t - \sin t) \end{pmatrix}$

$$\Rightarrow \quad Y_p = \begin{pmatrix} \frac{2}{5}(\sin t - 3\cos t) + \frac{1}{2}(5\cos t - \sin t) \\ \frac{3}{5}(\sin t - 3\cos t) + \frac{1}{2}(5\cos t - \sin t) \end{pmatrix}$$

$$= \begin{pmatrix} \frac{(25-12)}{10}\cos t + \frac{(-5+4)}{10}\sin t \\ \frac{(25-18)}{10}\cos t + \frac{(6-5)}{10}\sin t \end{pmatrix} = \begin{pmatrix} 1.3\cos t - .1\sin t \\ .7\cos t + .1\sin t \end{pmatrix}$$

Therefore

$$Y = Y_h + Y_p$$

$$= c_1 \begin{pmatrix} 1 \\ 1.5 \end{pmatrix} e^{-2t} + c_2 \begin{pmatrix} 1 \\ 1 \end{pmatrix} e^{-t}$$

$$+ \begin{pmatrix} 1.3\cos t - .1\sin t \\ .7\cos t + .1\sin t \end{pmatrix}$$

- So, we get the same solution in both methods.

Exercise: Find particular solution of

$$Y' = \begin{pmatrix} -3 & 1 \\ 1 & -3 \end{pmatrix} Y + \begin{pmatrix} -6 \\ 2 \end{pmatrix} e^{-2t}$$

using the method of variation of parameter.

Few Preliminaries:

- **Convergent Sequence:** A sequence of real numbers $\{a_n\}$ is said to converge to a real number l if for any given $\epsilon > 0$, there exists a positive integer $N = N(\epsilon)$ such that $|a_n - l| < \epsilon \quad \forall\, n \geq N$

- **Convergent Series:** A series of real numbers $\sum_{n=1}^{\infty} a_n$ is said to be convergent

if the sequence of its partial sum

$$S_1 = a_1, \quad S_2 = a_1 + a_2, \quad S_3 = a_1 + a_2 + a_3, \ldots \ldots$$
$$S_n = \sum_{i=1}^{n} a_i, \ldots$$ is convergent.

The limit of this sequence is called the sum of the series $\sum_{n=1}^{\infty} a_n$.

- **Sequence of functions:** A sequence of functions $\{f_n\}$ is said to converge to a function f on an interval I if for each $x \in I$, the sequence $\{f_n(x)\}$ converges to $f(x)$.

(i.e. for any given $\epsilon > 0$ and for $x \in I$, there exists a positive integer $N = N(\epsilon, x)$ such that $|f_n(x) - f(x)| < \epsilon \quad \forall\, n \geq N(\epsilon, x)$)

- <u>Series of functions:</u> A series of functions $\sum_{n=1}^{\infty} f_n$ is said to be convergent on an interval I if the sequence of its partial sums $S_1 = f_1$, $S_2 = f_1 + f_2$, ...

$$S_n = \sum_{i=1}^{n} f_i \quad \text{is convergent on } I.$$

The limit function of this sequence is called the sum of the series (of functions)

- <u>Power series:</u> A power series is a series of functions of the form $\sum_{m=0}^{\infty} a_m (x - x_0)^m$ ...... (1)

$f_n = (x - x_0)^n$

where $a_0, a_1, a_2, \dots$ are constants (coefficients) and x_0 is called the center of the series.

In particular if $x_0 = 0$, we get a power series of the form $\sum_{m=1}^{\infty} a_m x^m$

- Now if $x = x_0$, then the power series $\sum_{m=0}^{\infty} a_m (x-x_0)^m$ reduces to a_0 (a constant)

and so the power series (1) converges at $x = x_0$

In some cases, this may be the only value of x for which (1) converges.

If there are other values of x for which the series converges, these values form an interval called "the interval of convergence".

- The interval may be finite with midpoint x_0

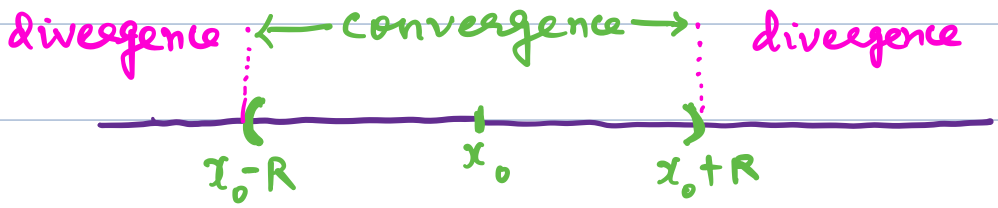
Then the power series (1) converges for all x in $(x_0 - R, x_0 + R)$

(ie in $\{x \in \mathbb{R} : |x - x_0| < R\}$)

and diverges for $|x - x_0| > R$

- The interval may also be infinite ie.

the series may Converge for all $x \in \mathbb{R}$



The quantity R is called the Radius of convergence.

- If the power series converges for all x, $R = \infty$ (i.e. $\frac{1}{R} = 0$)

- The radius of convergence can be determined from the coefficients of the series by the formulas:

(a) $$R = \frac{1}{\lim\limits_{n \to \infty} \left(|a_n|\right)^{1/n}}$$

(b) $$R = \frac{1}{\lim\limits_{n \to \infty} \left| \frac{a_{n+1}}{a_n} \right|}$$

provided the limits exist.

(If $R = 0$ (i.e. these limits are ∞) then the power series 1 converges only at the center x_0)

Existence of Power series solution:

If p, q and r in $y'' + p(x)y' + q(x)y = r(x)$ have power series representation at $x = x_0$, then every solution of the differential equation is differentiable in a neighborhood of x_0 and can be represented as power series in powers of $x - x_0$ with radius of convergence $R > 0$.

Ex: (1) $e^x = \sum_{m=0}^{\infty} \frac{1}{m!} x^m$

$$\frac{a_{m+1}}{a_m} = \frac{\frac{1}{(m+1)!}}{\frac{1}{m!}} = \frac{m!}{(m+1)!} = \frac{1}{m+1}$$

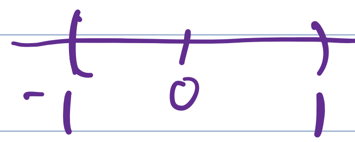
$$\lim_{m \to \infty} \left| \frac{a_{m+1}}{a_m} \right| = \lim_{m \to \infty} \frac{1}{m+1} = 0$$

So, $R = \frac{1}{\lim_{m \to \infty} \left| \frac{a_{m+1}}{a_m} \right|} = \boxed{\infty}$

(2) $$\frac{1}{1-x} = 1 + x + x^2 + \cdots\cdots$$

$$\lim_{m\to\infty}\left|\frac{a_{m+1}}{a_m}\right| = \lim_{m\to\infty}\left|\frac{1}{1}\right| = 1$$

So, $$R = \frac{1}{\lim_{m\to\infty}\left|\frac{a_{m+1}}{a_m}\right|} = \frac{1}{1} = \boxed{1}$$



(3) $$\sum_{m=0}^{\infty} m!\, x^m$$

$$\lim_{m\to\infty}\left|\frac{a_{m+1}}{a_m}\right| = \lim_{m\to\infty}\frac{(m+1)!}{m!}$$

$$= \lim_{m\to\infty}(m+1) = \infty$$

So, $$R = \frac{1}{\lim_{m\to\infty}\left|\frac{a_{m+1}}{a_m}\right|} = \frac{1}{\infty} = \boxed{0}$$

Examples of Power Series:

Ex: Taylor Series: $f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(x_0)}{n!} (x-x_0)^n$

Ex: $\frac{1}{1-x} = 1 + x + x^2 + x^3 + \cdots\cdots$ (geometric series) $(|x| < 1)$

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots\cdots$$

$$\cos x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!}$$

$$\sin x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!}$$

Ex: $e^x = 1 + x + \frac{1}{2!}x^2 + \frac{1}{3!}x^3 + \cdots\cdots$

(centered around zero)

$$e^x = e + e(x-1) + e\frac{1}{2!}(x-1)^2 + e\frac{1}{3!}(x-1)^3 + \cdots$$

(centered around 1)

Series solutions of ODE:

Power Series Method:

$$y' - y = 0$$

We look for a solution of the form:

$$Y = \sum_{m=0}^{\infty} a_m x^m$$

$$y' = \sum_{m=1}^{\infty} m a_m x^{m-1}$$

Substituting in the equation: $y' - y = 0$

$$\left(a_1 + 2a_2 x + 3a_3 x^2 + \cdots\right) - \left(a_0 + a_1 x + a_2 x^2 + \cdots\right) = 0$$

$$\Rightarrow (a_1 - a_0) + (2a_2 - a_1)x + (3a_3 - a_2)x^2 + \cdots = 0$$

$$\Rightarrow \quad a_1 - a_0 = 0 \quad \Rightarrow \quad a_1 = a_0$$

$$2a_2 - a_1 = 0 \quad \Rightarrow 2a_2 = a_1 = a_0 \Rightarrow a_2 = \frac{1}{2} a_0$$

$$3a_3 - a_2 = 0 \Rightarrow 3a_3 = a_2 \Rightarrow a_3 = \frac{1}{3.2} a_0$$

$$\Rightarrow a_3 = \frac{1}{3!} a_0$$

$$\vdots$$

In general $a_k = \frac{1}{k!} a_0$

So, $Y = a_0 + a_0 x + \frac{1}{2} a_0 x^2 + \frac{1}{3!} a_0 x^3 + \cdots$

$$= a_0 \left(1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots\cdots \right)$$

$$= a_0 e^x$$